

X-ray analyses on a painted wood panel

Unknown pigments

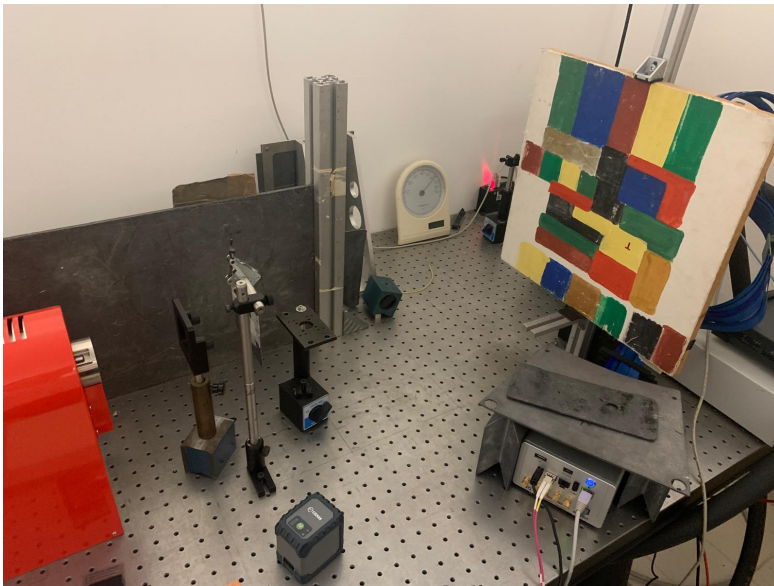
Linda Biasci

02/07/2026

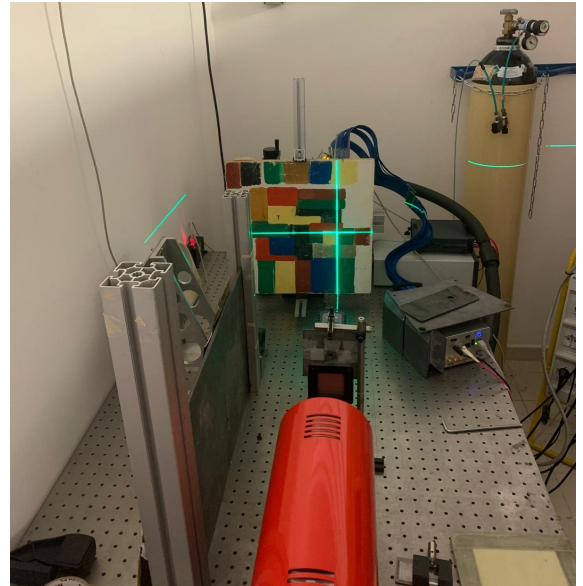
Introduction

X-ray imaging and transmission analyses are performed on different areas of a painted wood panel with Timepix4 and an Amptek detector, aiming to recognise elements or substances by which pigments are constituted, relying on K-edge analyses.

Experimental setup



(a)



(b)

Figure 1: Experimental setup; figure (a), from left to right: X-ray tube, filtration, collimator, wood painting, (behind the panel) encased Timepix4 immersed in nitrogen, in thermal contact with the heat exchanger and in electrical contact with the read-out system, shielded by lead plates; in figure (b), the laser level is employed to focus the measure on a panel's ROI

In both experimental setups, the employed X-ray detector is placed behind the painted board (fixed on the table with a metallic support), at a distance varying in the range $[5\text{ cm}, 10\text{ cm}]$. The distance between the source and the Timepix4 ASIC is approximately 80 cm and the distance between the source and the Amptek detector is approximately 130 cm . The board position is modified manually before each acquisition, in order to measure X-ray transmission of different pigments and sections of the whole panel. For each measurement, such position is indeed regulated using a CM-G02 Laser Level (Cigman), spotting different areas in the picture.

Source

The employed source is a micro-focus X-ray tube (Hamamatsu), characterised by:

- tungsten anode in transmission mode
- energy range: $kVp = 120\text{ keV}$ (only in Timepix4 acquisitions) and $kVp = 60\text{ keV}$
- anode current: $I = 10\text{ }\mu\text{A}$ (Timepix4 acquisitions) and $I = 6\text{ }\mu\text{A}$ (Amptek acquisitions)
- filtration: two aluminum filters of width 1 mm each and four copper filters of width $250\text{ }\mu\text{m}$ each for 120 kVp measurements, two aluminum filters of width 1 mm each for 60 kVp measurements
- acquisition time: 1000 s (Timepix4 acquisitions, 10 runs of 100 s each) and 100 s (Amptek acquisitions)
- collimator: a lead slab of thickness 2 mm , with a square hole of size 9 mm (Timepix4 acquisitions) and a tungsten disk of thickness 2 mm , with a hole diameter of 1 mm (Amptek acquisitions)

Sample

The analysed sample is a test painting (a “mock up”): unknown pigments of different colours and materials are spread on a white substrate, painted on a thin canvas placed on a wood panel of thickness $1.8 \pm 0.1\text{ cm}$. The test picture dimensions are $39.5 \times 39.5\text{ cm}^2$.

Detection systems

In the first measurements set, Timepix4 is used with a CdTe detector, and the detection system is characterised by:

- CdTe thickness: 1 mm
- expected energy resolution: $6\text{ keV} < FWHM < 10\text{ keV}$
- pixel area: $55 \times 55\text{ }\mu\text{m}^2$
- detector area: 448×512 pixels, i.e. total detector area of 6.94 cm^2
- cooling system: Timepix4 is placed in thermal contact with a copper heat exchanger and a chiller liquid continuously flows inside of it, and it is inserted in a closed case inside of which nitrogen flows at approximately 1 l/min
- readout system: SPDR4 (Speedy Pixel Detector Readout), which can support a maximum acquisition rate of approximately 10 million events per second

In the second measurements set, the employed detector is a XR-100CdTe X-Ray and Gamma Ray Detector (Amptek), characterised by:

- thickness: 1 mm
- efficiency $> 90\%$ in the range $[5\text{ keV}, 70\text{ keV}]$
- energy resolution: $FWHM < 1.5\text{ keV}$
- detector area: $5 \times 5\text{ cm}^2$

Methods

Timepix4 analyses

The X-ray spectrum is firstly acquired without the sample (flat field acquisition with $kVp = 120\text{ keV}$ and with $kVp = 60\text{ keV}$) in order to obtain the I_0 signals for further analysis; then, the panel is inserted and the I spectrum is acquired on various sections (four acquisitions with $kVp = 120\text{ keV}$ and four acquisitions with $kVp = 60\text{ keV}$). The applied calibration parameters were derived using a combination of synchrotron radiation (energy calibration up to 38 keV), fluorescence photons (molybdenum, gadolinium and lead for higher energy calibration) and test pulses (calibration with electric signals), and refined to correct non-uniformity effects on the two different ASIC’s half-matrices.

A Timepix4 acquisition in data-driven mode allows to obtain the position (x, y) of each occurring event, the Time of Arrival and the Time over Threshold (ToA, ToT) of events with a fixed threshold of 700 e^- . Information gained through position, ToT and ToA is combined to perform clustering of detected events, i.e. to group hits in clusters on

pixels, and to measure the energy of incident radiation. For the purposes of this work, energy information on counts and clustering of events are only taken into consideration.

For each 100 s-lasting run, acquired energy information on counts is clustered and processed to obtain nine different sets of, respectively, 150 and 100 subsequent stacked images (TIFF format), corresponding to energy slicing in the range $[0 \text{ keV}, 150 \text{ keV}]$ and $[0 \text{ keV}, 100 \text{ keV}]$ (the actual signals extending up to 120 keV and 60 keV , respectively) with an energy bin width of 1 keV . Each different set of stacks is associated to a fixed cluster size: $\{1, 2, 3, 4, 5, 6, 7, 8\}$ and “all” (i.e. superposition of signals from all cluster sizes). To avoid undesired effects due to the time variation of clustered events percentages, only signals with cluster size “all” are taken into account.

For each acquired panel section, the respective I_0 file is compared to the I file, so to derive total attenuation spectra and extrapolate information about the expected K-edges. Therefore, for cluster size “all”, stacks are processed in a final $-\ln(I/I_0)$ spectrum.

To enhance K-edge visibility, subtraction analyses are performed on each pigment’s total attenuation clustered data. Having selected an energy bin width of 1 keV but being the Timepix4 experimental resolution characterised at least by a $FWHM = 6 \text{ keV}$, total attenuation spectra are also over-stacked (summed up) to increase the actual bin width (hence, counts statistics) and potentially improve the quality of differential Kes results. K-edge subtraction analyses (as well as attenuation analyses) are performed with energy bin widths $\{1, 5\} \text{ keV}$ for 120 kVp and $\{1, 2\} \text{ keV}$ for 60 kVp .

Amptek analyses

A preliminary instrumental calibration is required before starting transmission acquisitions. At first order approximation, keV to channels conversion is assumed to be linear (approximately, a conversion parameter $[ch]/[keV] \sim 7.5$ is expected): calibration is performed with a linear fit on the incident spectrum, i.e. with a model function $y = c \cdot x + a$, being c the conversion parameter (from keV to channels) and a a possible calibration offset. The such previously obtained fit results (based on 120 kVp flat field acquisition and known energy positions of tungsten fluorescence peaks) are: calibration factor $c = 7.673 \pm 0.151$, offset $a = 10.996 \pm 9.604$, chi square / dof = $1.063/2$. A less precise calibration is computed again, comparing the latest 60 kVp flat field Amptek acquisition to a 60 kVp flat field acquired with Timepix4 in the same experimental conditions. Obtained fit results are compatible with (although, as expected, apparently less accurate than) the previous ones: calibration factor $c = 7.434 \pm 0.367$, offset $a = 15.300 \pm 12.889$, chi square / dof = $33.527/3$. In conclusion, calibration parameters yielded from 120 kVp spectrum with tungsten peaks are kept in the presented acquisitions.

Acquired data is saved and converted in a `csv` file. Radiation emitted from the source and transmitted through the different parts of the panel is detected and respective signals (counts spectra) are visualised.

Data are then analysed to produce total attenuation spectra and possibly spot K-edges of materials present in pigments and consequently derive pigments thickness. Hence, the function $-\ln(I/I_0) = \sum \mu(E) \cdot x$ is calculated and plotted, being I the signal of radiation transmitted by the sample and I_0 the background or flat field signal, that could either be the incident spectrum or the signal that best describes a pure wood spectrum. A moving average smoothing via the `Python` function `numpy.convolve` with a pre-defined filter is applied to data, in order to reduce stochastic noise and improve the visualisation of experimental data. For each acquired pigment or section, two smoothed total attenuation spectra are computed, using both the 60 kVp flat field and the wood spectrum (obtained averaging over four spatially different acquisitions on panel’s areas uncovered by pigments) as I_0 signals.

Total attenuation spectra of different pigments are plotted and saved in `csv` files: attenuation values are extracted in correspondence of different positions of interest (e.g. K-edges of materials) and tabulated $\mu(E)$ are used to derive pigments or wood thicknesses. To estimate pigments thickness, the averaged wood signal (which actually contains canvas and white substrate signals, too) is taken as the I_0 signal, whereas to estimate wood thickness contribution of the white pigment substrate is assumed negligible in attenuation because of its thinness.

Results: Amptek

Results of X-ray transmission acquired with the Amptek detector are reported first, as they contributed to the interpretation of data acquired with Timepix4.

Flat field and wood

An average over the whole four wood signals is computed in order to generate a more uniform I_0 signal for further analyses, plotted in orange in the figure below (`60kV_6uA_100s_2mmAl_meanwood`).

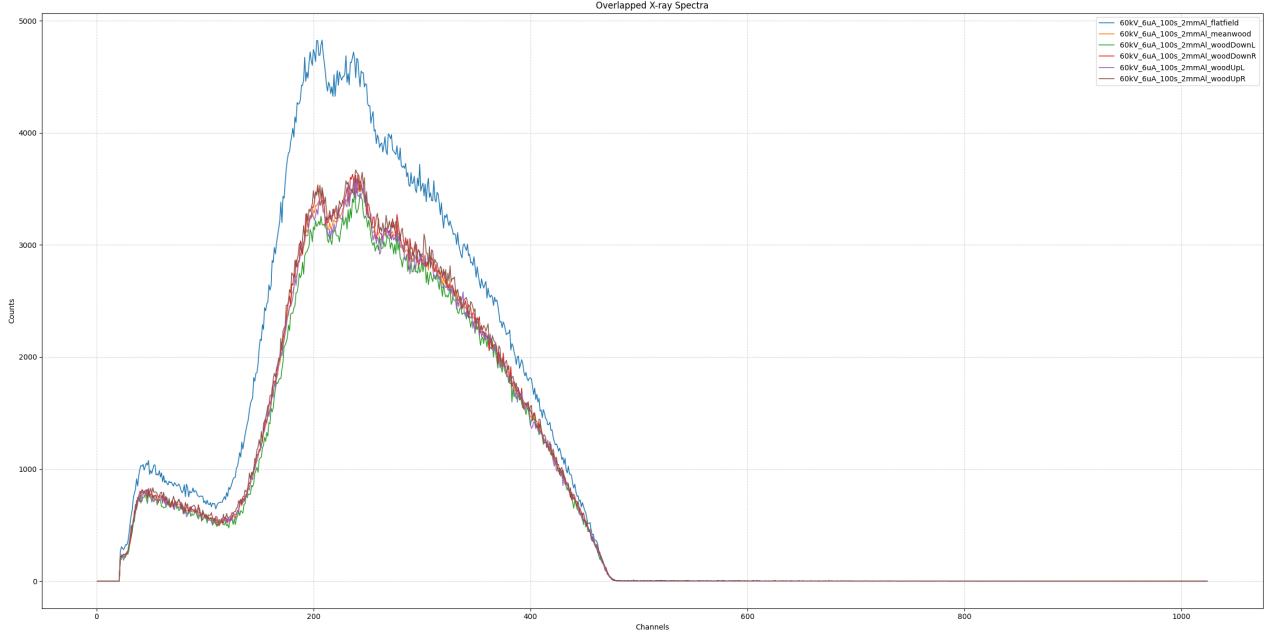


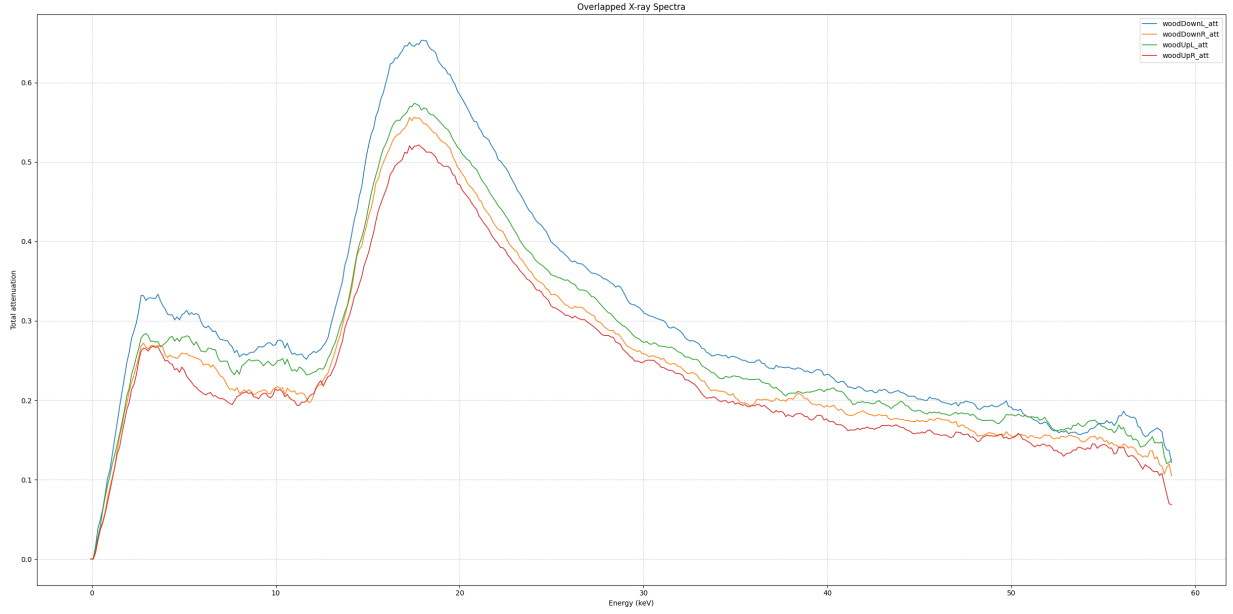
Figure 2: Measured spectra of incident radiation (flat field at 60 kVp , blue line in the plot) and radiation transmitted through the four pigment-less corners of the wood panel; energy channels (from 1 to 1024) are represented on horizontal axis, photon counts are represented on vertical axis

Wood thickness is estimated basing on the mean attenuation signal: total attenuation values are extracted from the `csv` file at different energies and thickness measurements are derived dividing such attenuations by the tabulated (roughly averaged) coefficient values $\mu_{wood}(E)$. The actual thickness estimation is taken as the mean of five measurements obtained as described, and their standard deviation is taken as the measurement uncertainty. The yielded width is compared to the directly measured one, and discrepancies are to be attributed to the rough estimation of the linear attenuation coefficient of wood, as the precise density of the considered wood is not known.

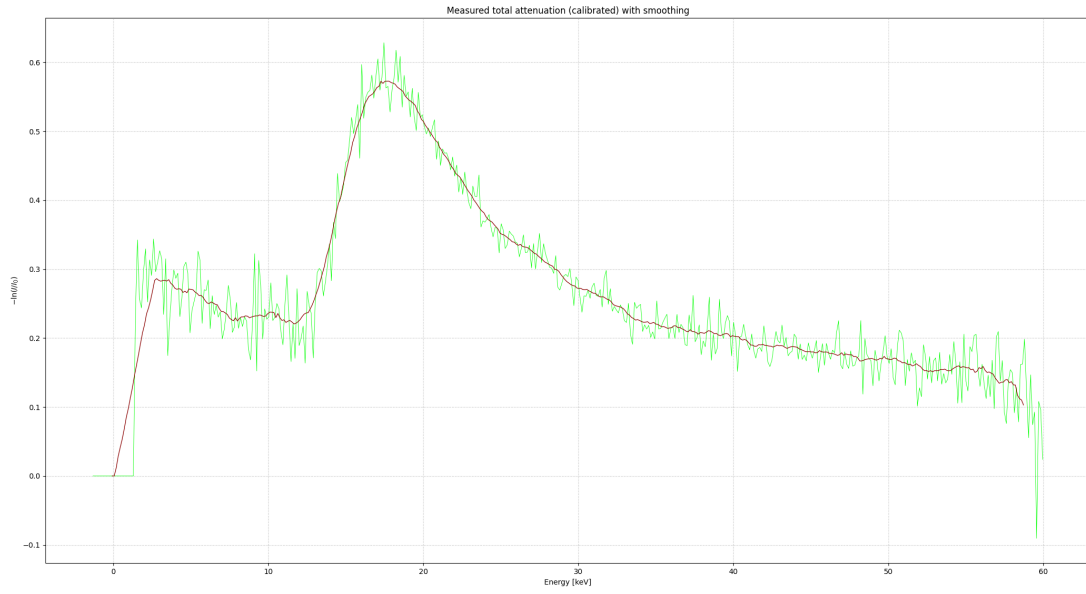
energy value [keV]	25.088	30.041	35.124	45.029	53.109
measured attenuation ($\mu(E) \cdot x$)	0.350	0.272	0.220	0.180	0.153
derived wood thickness [cm]	1.460	1.511	1.466	1.499	1.453

Table 1: Wood thickness estimation from attenuation measurements

The final (average) wood thickness is therefore $x_{wood} = 1.48 \pm 0.03$ cm , to be compared with the measured width $x_{wood} = 1.8 \pm 0.1$ cm . A measurement of the panel's mass (i.e. wood density) is required if a more accurate wood thickness estimation is requested.



(a) Attenuation signals of the wood panel's four pigment-less corners, taking the 60 kVp flat field as I_0



(b) Raw (in green) and smoothed (in red) attenuation signal of the average wood counts, taking the 60 kVp flat field as I_0

Figure 3: Plots of $-\ln(I/I_0)$ for the wood panel (also containing minor canvas and white pigment contributions)

Pigments

The reported attenuation results are obtained using pigment signals for I and the average wood signal for I_0 (results with flat field employed as I_0 are not reported, as they confirm what obtained with wood but they allow a less accurate estimation of thicknesses). The following pigments on the wood panel are analysed:

- gold (one acquisition, localised where the pigment appears to be at its maximum thickness)
- bright green (one acquisition)
- bright red (two acquisitions on two different spots to verify they comprehend the same kind of pigment)
- dark green (one acquisition)
- dark red (one acquisition)
- blue (one acquisition, localised where the pigment appears to be at its maximum thickness)
- pastel yellow (one acquisition)

Almost every analysed pigment shows a K-edge characteristic of barium and some pigments show a K-edge characteristic of zinc: these signals could be attributed to barium sulphate and possibly (in a smaller concentration) to zinc sulphide, as these compounds appear as a white powder, which could have been used in the white pigment that covers the panel's surface, as a substrate for the pigments of interest.

Amptek does not allow to detect any evident trace of other expected elements (e.g. cadmium in red or green pigments): such absence might be attributed to the fact that the wood and the substrate, with their relevant thicknesses, cover all pigments attenuation signals (being these latter undeniably thin, even when observed with naked eye).

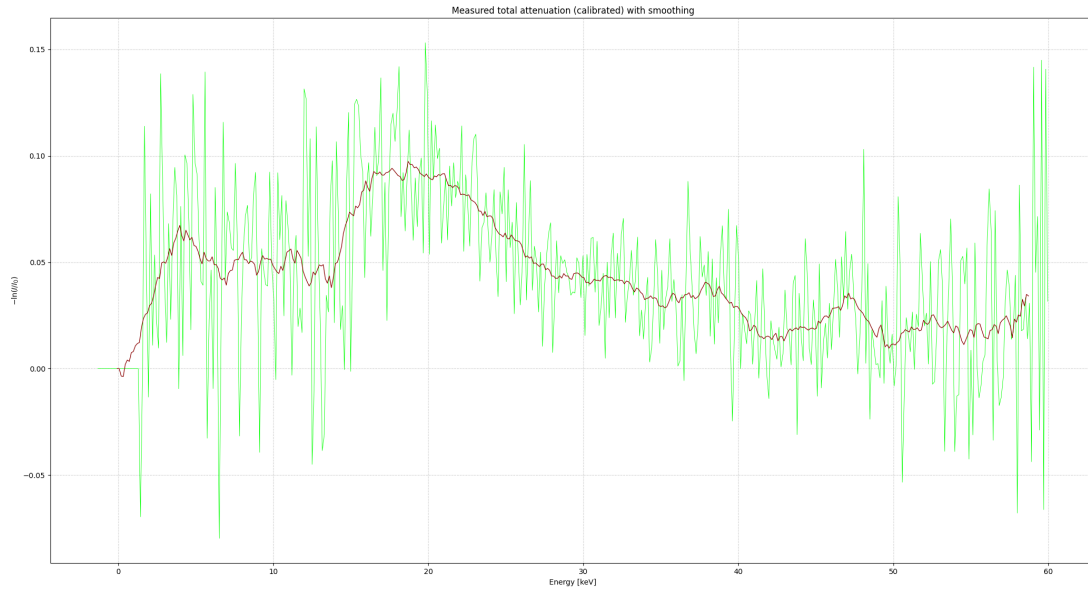


Figure 4: Total attenuation signal on a golden ROI of the panel, showing no evident barium and zinc K-edges

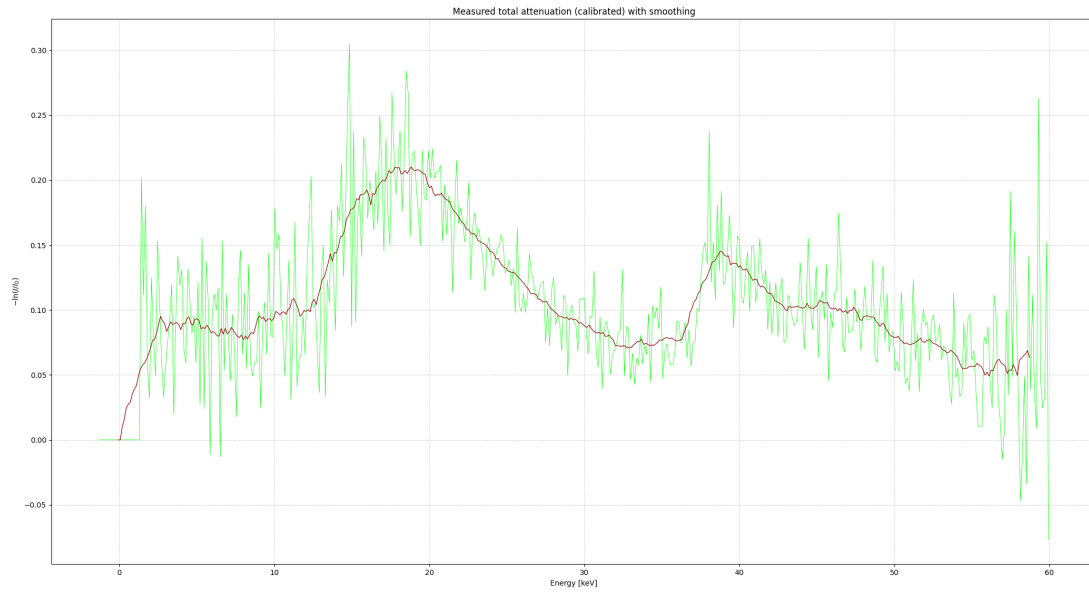
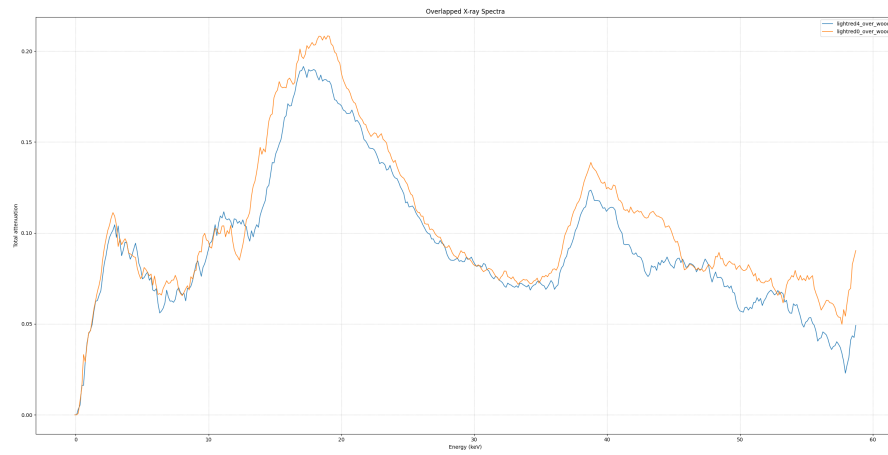
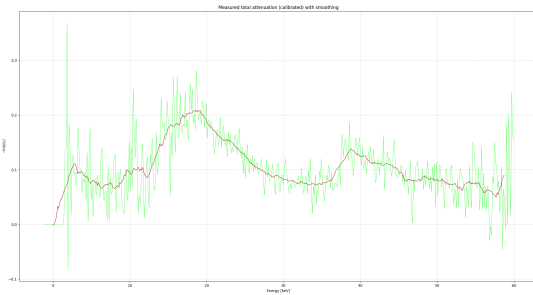


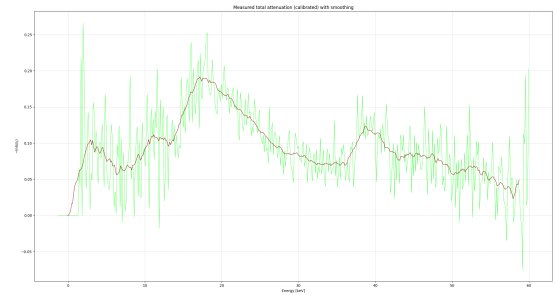
Figure 5: Total attenuation signal on a bright green ROI of the panel, showing a strong barium K-edge and a weak zinc K-edge



Comparison between two different bright red ROIs, allowing to be identified as the same pigment



Total attenuation, first bright red ROI of the panel



Total attenuation, second bright red ROI of the panel

Figure 6: Total attenuation signal on two different bright red ROIs of the panel, showing both a strong barium K-edge and an evident zinc K-edge

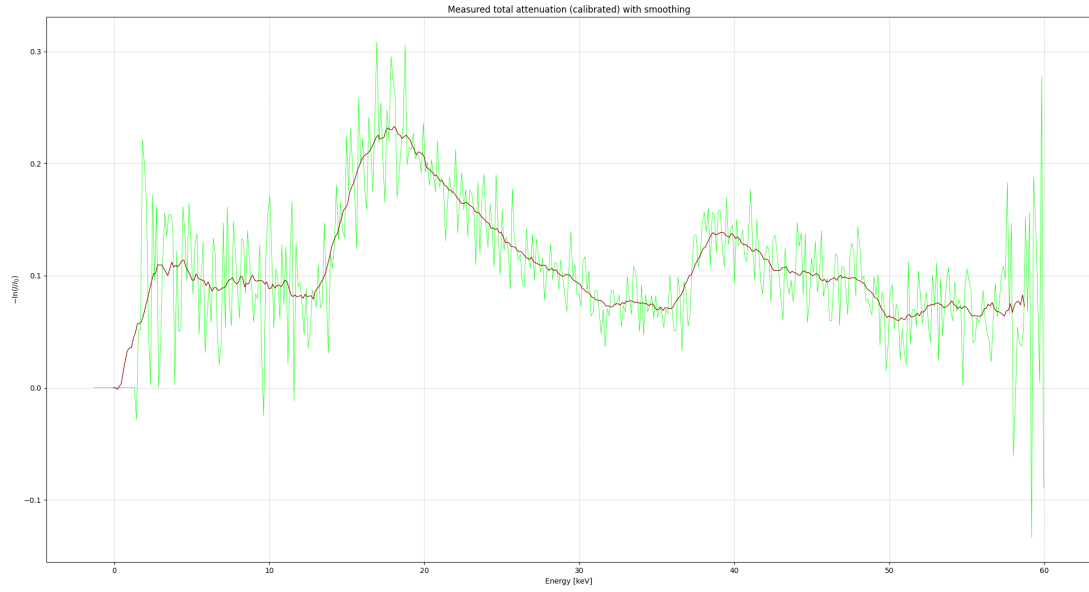


Figure 7: Total attenuation signal on a dark green ROI of the panel, showing a strong barium K-edge and no evident zinc K-edge

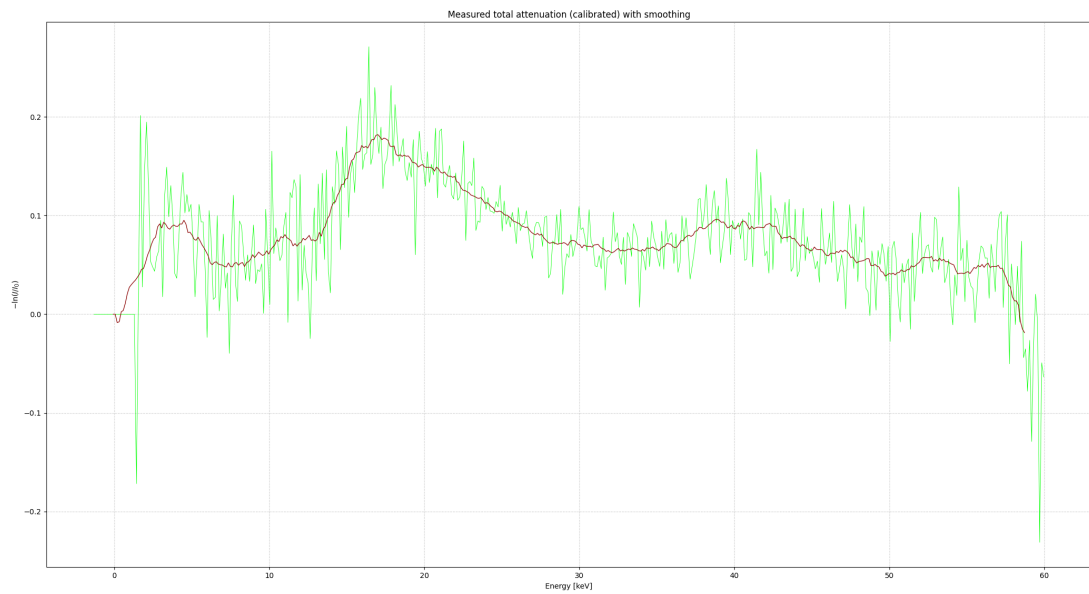


Figure 8: Total attenuation signal on a dark red ROI of the panel, showing weak barium and zinc K-edges

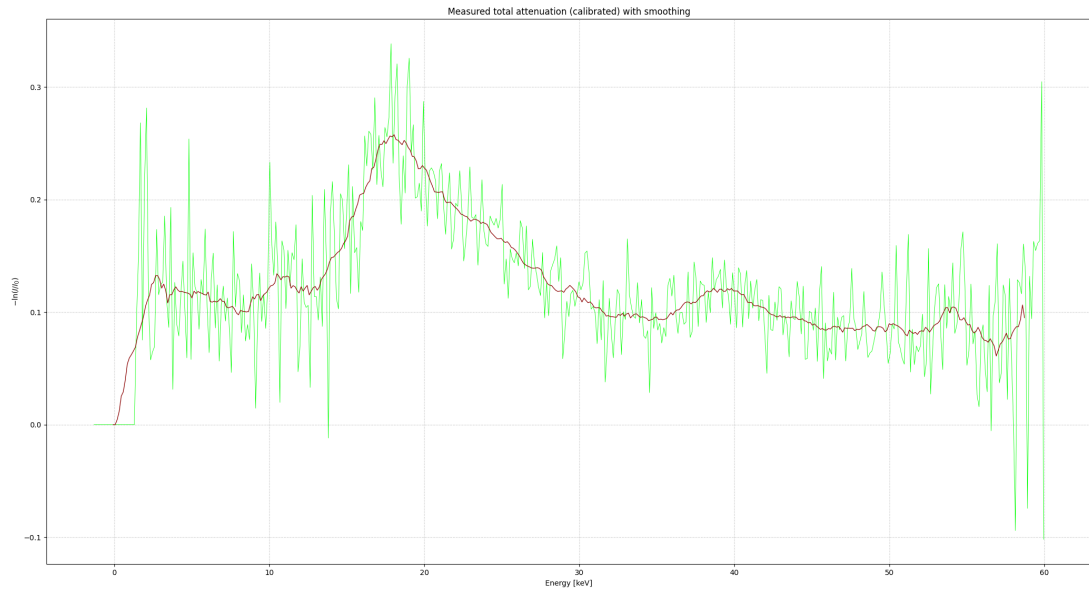


Figure 9: Total attenuation signal on a blue ROI of the panel, showing evident barium and zinc K-edges

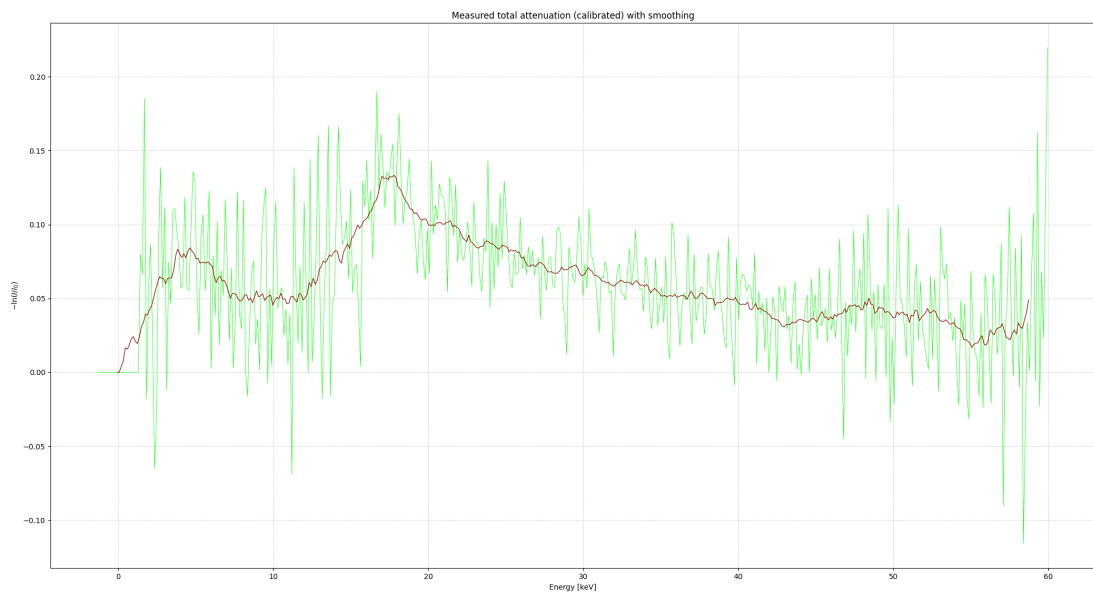


Figure 10: Total attenuation signal on a yellow ROI of the panel, not showing any K-edge

Substrate thickness estimation

Neglecting barium contribution in correspondence of zinc's K-edge and zinc contribution in correspondence of barium's K-edge, thicknesses of the assumed zinc sulphide and barium sulphate are estimated from attenuation signals of pigments.

Zinc K-edge is expected at $E_K = 9.659 \text{ keV}$ and its correspondent attenuation coefficient is $\mu_{Zn} = 1808.9 \text{ cm}^{-1}$. Averaging over five acquisitions (presented in Table 2), the assumed zinc sulphide is expected to have an approximate width of $x_{ZnS} \simeq 0.509 \pm 0.103 \text{ }\mu\text{m}$.

pigment	bright green	bright red (1)	bright red (2)	blue	dark red
measured attenuation ($\mu(E) \cdot x$)	$9.33 \cdot 10^{-2}$	$9.98 \cdot 10^{-2}$	$8.65 \cdot 10^{-2}$	$11.62 \cdot 10^{-2}$	$6.42 \cdot 10^{-2}$
derived thickness [μm]	$5.16 \cdot 10^{-1}$	$5.52 \cdot 10^{-1}$	$4.78 \cdot 10^{-1}$	$6.43 \cdot 10^{-1}$	$3.55 \cdot 10^{-1}$

Table 2: ZnS thickness estimation from attenuation measurements on thin pigments (over wood)

Barium K-edge is expected at $E_K = 37.441 \text{ keV}$ and its correspondent attenuation coefficient is $\mu_{Zn} = 102.2 \text{ cm}^{-1}$. Averaging over six acquisitions (presented in Table 3), the assumed barium sulphate is expected to have an approximate width of $x_{BaSO_4} \simeq 10.207 \pm 1.137 \text{ }\mu\text{m}$.

pigment	dark green	bright green	bright red (1)	bright red (2)	blue	dark red
measured attenuation ($\mu(E) \cdot x$)	0.110	0.115	0.112	0.097	0.108	0.084
derived thickness [μm]	10.750	11.232	10.932	9.509	10.597	8.220

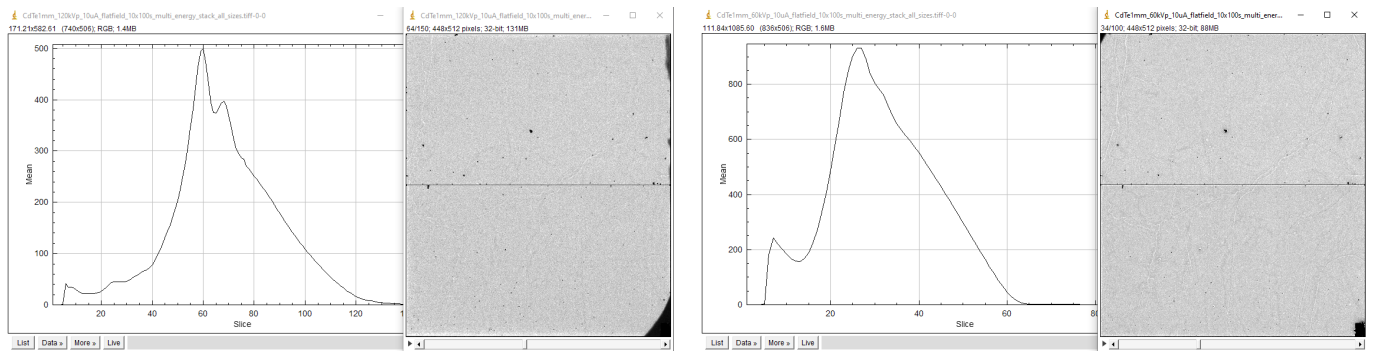
Table 3: $BaSO_4$ thickness estimation from attenuation measurements on thin pigments (over wood)

In conclusion, the white substrate can be assumed to have a width in the range $[10 \text{ }\mu\text{m}, 20 \text{ }\mu\text{m}]$, which does not allow to derive the thickness of pure pigments - also because no other K-edge value can be identified, except for barium and, with less accuracy, zinc. Tests have been made extracting attenuation signals far from what are expected to be the barium and zinc K-edges, but no reliable results have been obtained.

Results: Timepix4

Results obtained from Timepix4 imaging are reported for each analysed zone (or ROI, comprehensive of two or three different pigments). Only data with cluster size "all" is taken into account.

Flat field



Flat field counts, $kVp = 120 \text{ keV}$

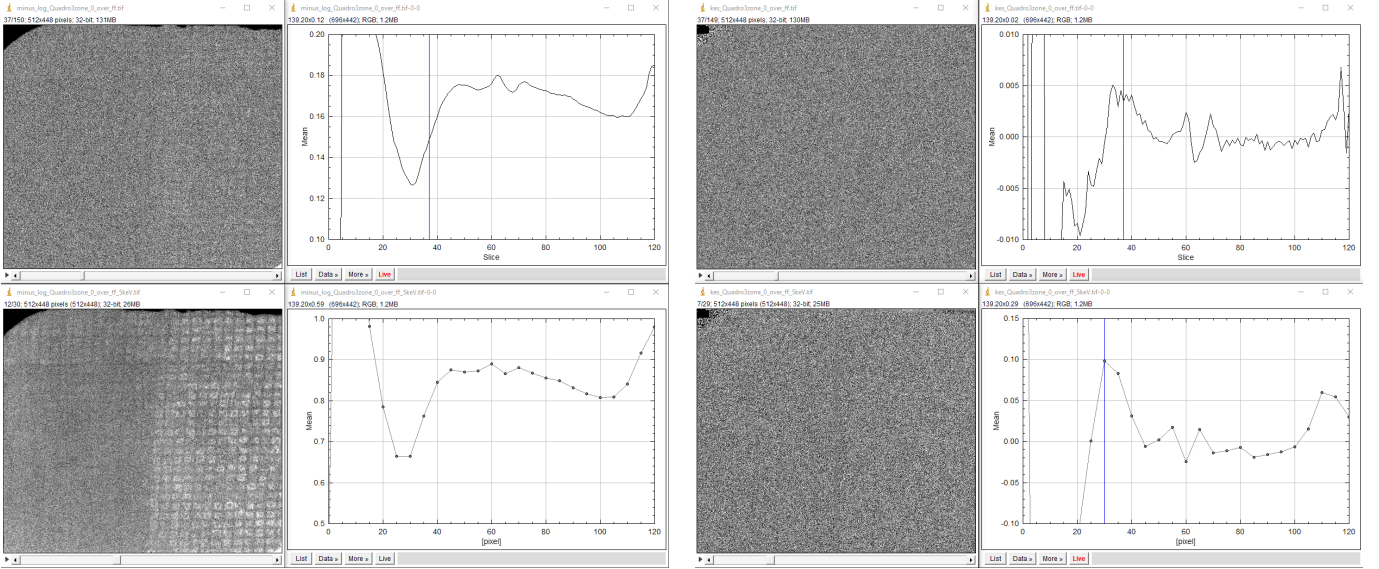
Flat field counts, $kVp = 60 \text{ keV}$

Figure 11: Detected counts with Timepix4 for flat field at two different source's kVp

Zone 0, 120 kVp and 60 kVp



Figure 12: Zone “0” is analysed both with 120 kVp and 60 kVp

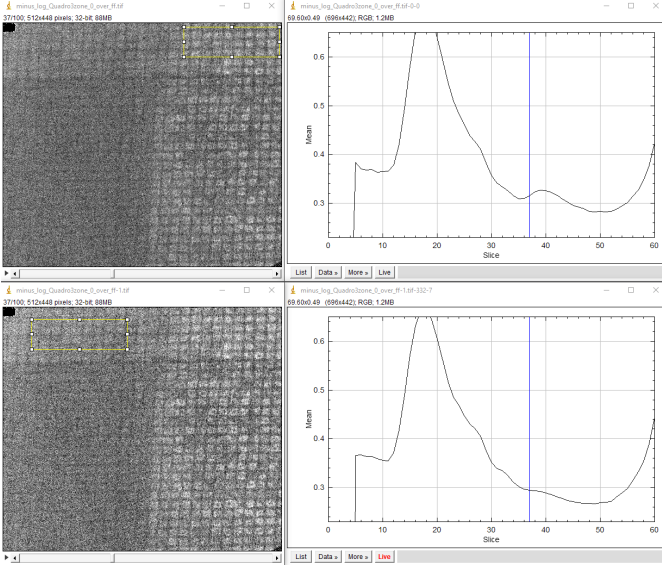


(a) Total attenuation, $kVp = 120 \text{ keV}$, with binwidth 1 keV (upper picture) and 5 keV (lower picture)

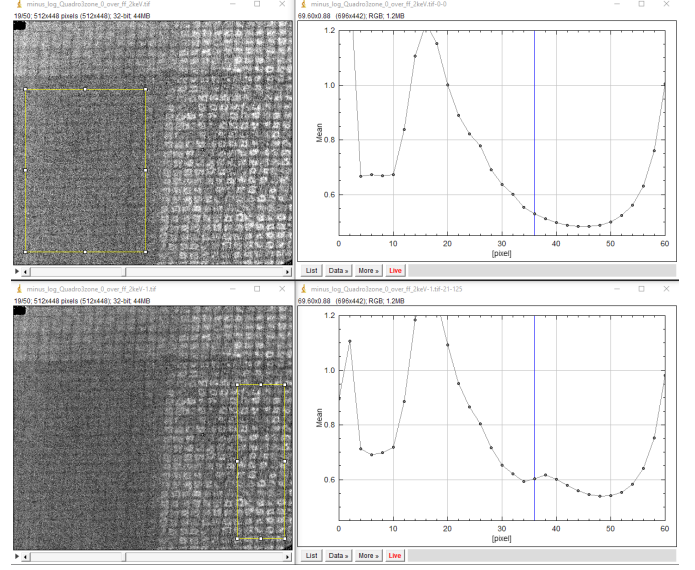
(b) K-edge subtraction, $kVp = 120 \text{ keV}$, with binwidth 1 keV (upper picture) and 5 keV (lower picture)

Figure 13: $-\ln(I/I_0)$ and Kes images for ROI 0, 120 kVp

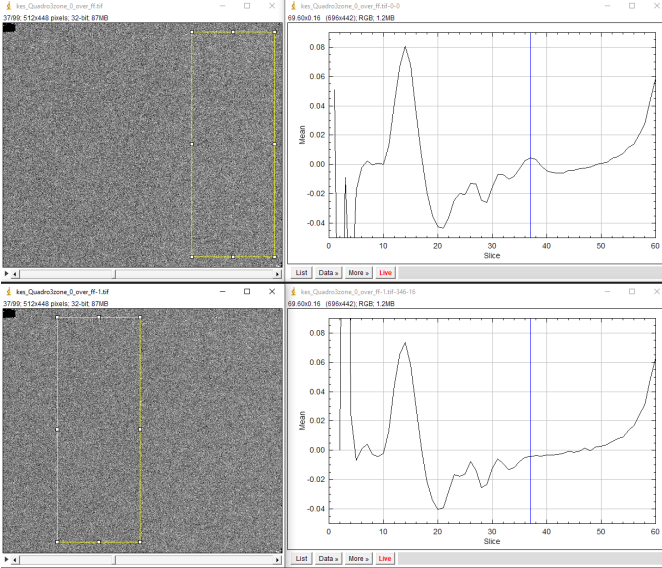
A K-edge appears around 37 keV , which might be attributed to the presence of barium. From total attenuation images, the yellow pigment (on the lower-left side) is expected to be more attenuating than the green one (lower-right side) and the red (upper) one. As no heavy element can be found in the yellow pigment with such a resolution and acquisition time, this latter can be deduced to be thicker than the others. Moreover, there is no trace of the hidden drawing that can be seen in Figure 12 (underneath the yellow pigment).



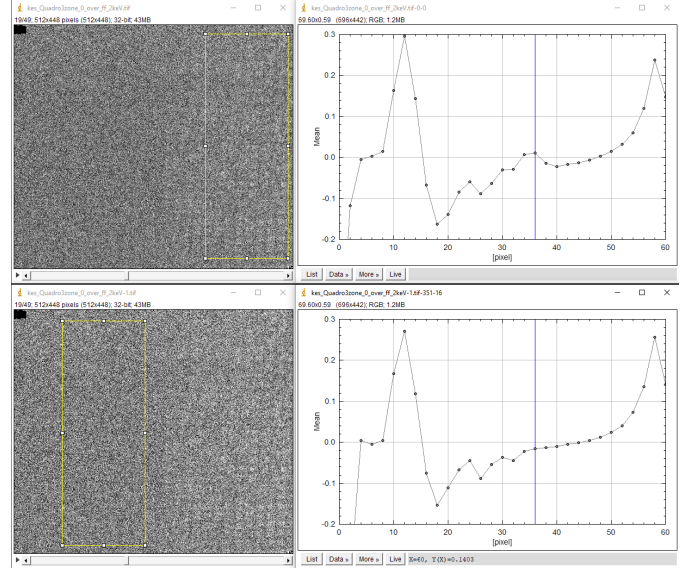
(a) Total attenuation, $kVp = 60 \text{ keV}$, with binwidth 1 keV



(b) Total attenuation, $kVp = 60 \text{ keV}$, with binwidth 2 keV



(c) K-edge subtraction, $kVp = 60 \text{ keV}$, with binwidth 1 keV



(d) K-edge subtraction, $kVp = 60 \text{ keV}$, with binwidth 2 keV

Figure 14: $-\ln(I/I_0)$ and Kes images for ROI 0, 60 kVp

As it can be seen in Figure 14 (a), the ROI of expected superposition of green and red pigments (upper picture) shows a K-edge at 37 keV , unlike the ROI of expected superposition of yellow and red pigments (lower picture); from Figure 14 (b) it can be deduced that this behaviour is due to the yellow pigment (upper picture) attenuating more than the green pigment (lower picture): indeed, no barium K-edge can be seen in the yellow ROI. K-edge imaging confirms this, as it can be seen in Figure 14 (c) and (d): no peak at 37 keV is displayed in the two lower pictures, mostly centred on the yellow pigment.

Zone 1, 120 kVp

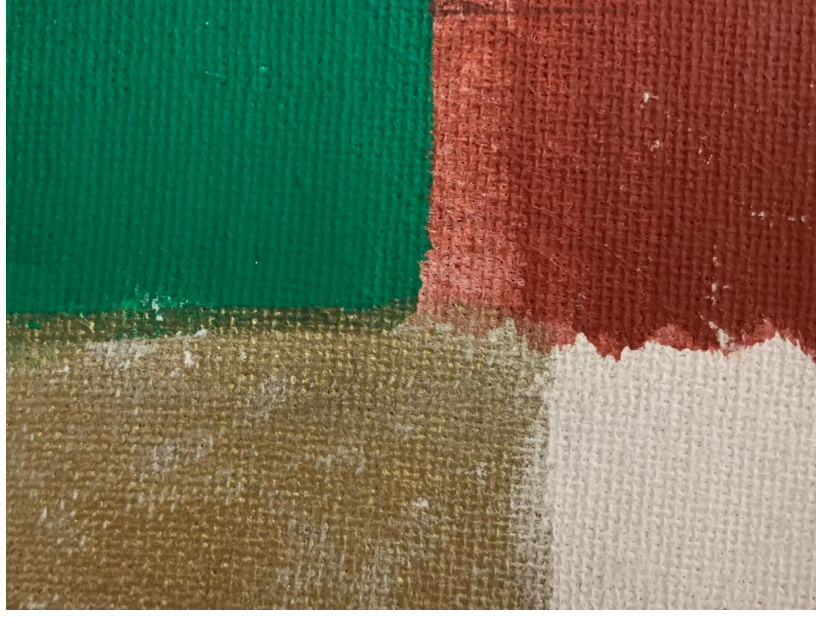
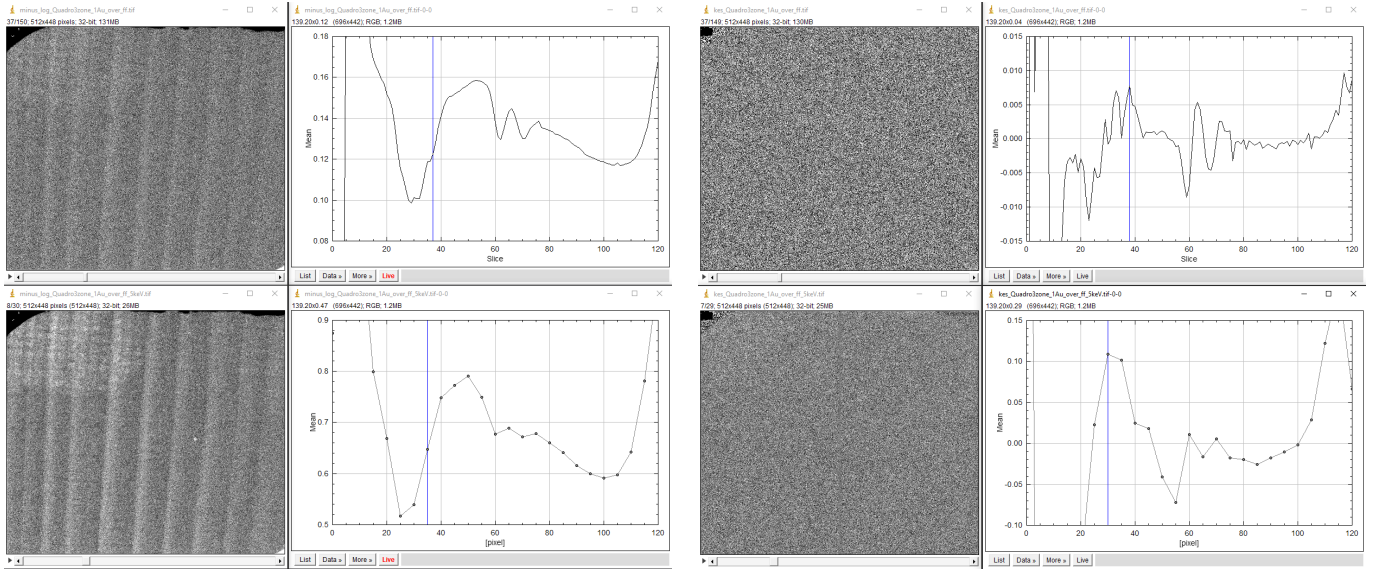


Figure 15: Zone “1” is analysed at 120 kVp (since gold presence is expected)



(a) Total attenuation, $kVp = 120 \text{ keV}$, with binwidth 1 keV (upper picture) and 5 keV (lower picture)

(b) K-edge subtraction, $kVp = 120 \text{ keV}$, with binwidth 1 keV (upper picture) and 5 keV (lower picture)

Figure 16: $-\ln(I/I_0)$ and Kes images for ROI 1, 120 kVp

No presence of gold can be confirmed, as no K-edge or peak in Kes images appears at $E_K = 80.725 \text{ keV}$. This might be due to the small thickness of the pigment compared to the expected thickness of the barium sulphate substrate and, most of all, to the panel width. Wood grain is indeed clearly visible in total attenuation stacks, whereas the three pigments present in the zone are almost indistinguishable, though they all show a K-edge around 37 keV .

Zone 2, 60 kVp

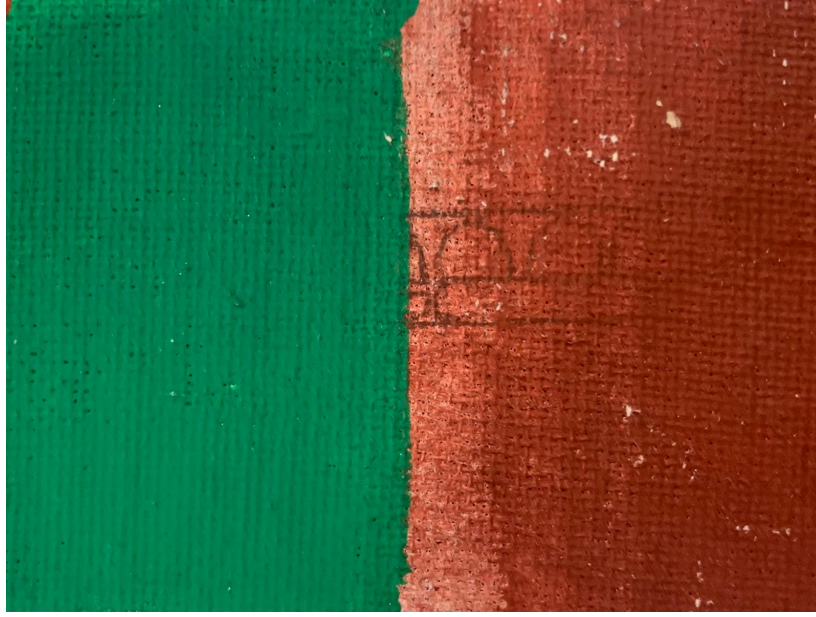
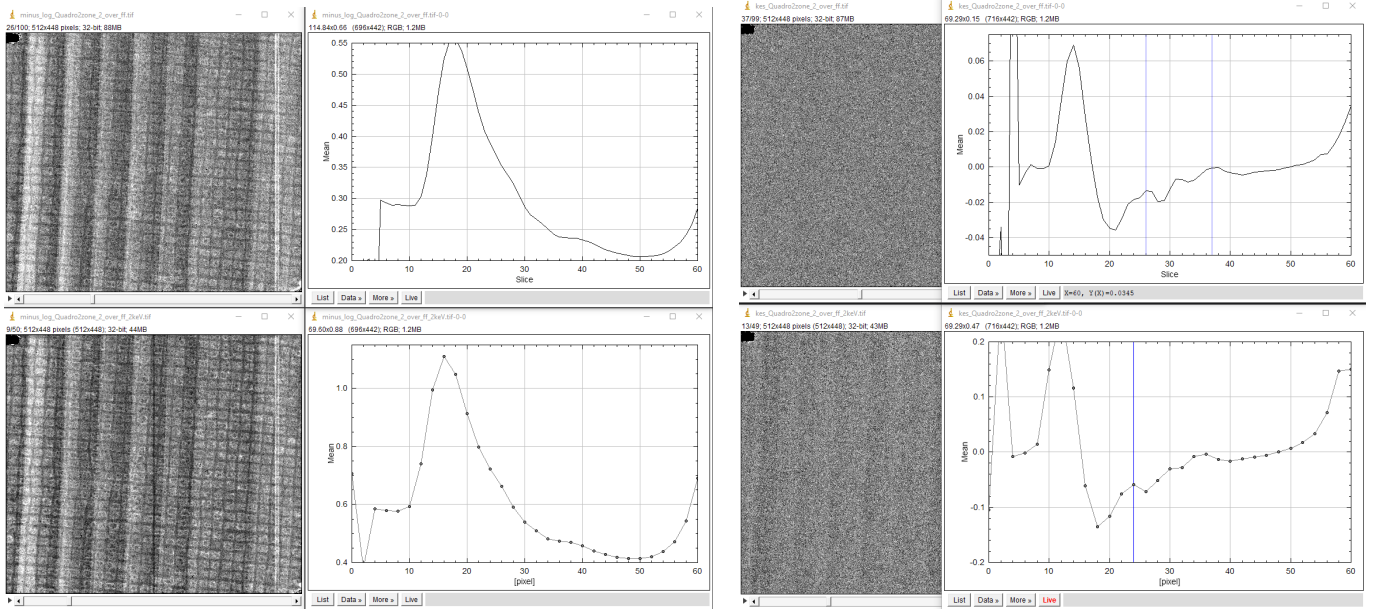


Figure 17: Zone “2” is analysed at 60 kVp



(a) Total attenuation, $kVp = 60$ keV , with binwidth 1 keV (upper picture) and 2 keV (lower picture)

(b) K-edge subtraction, $kVp = 60$ keV , with binwidth 1 keV (upper picture) and 2 keV (lower picture)

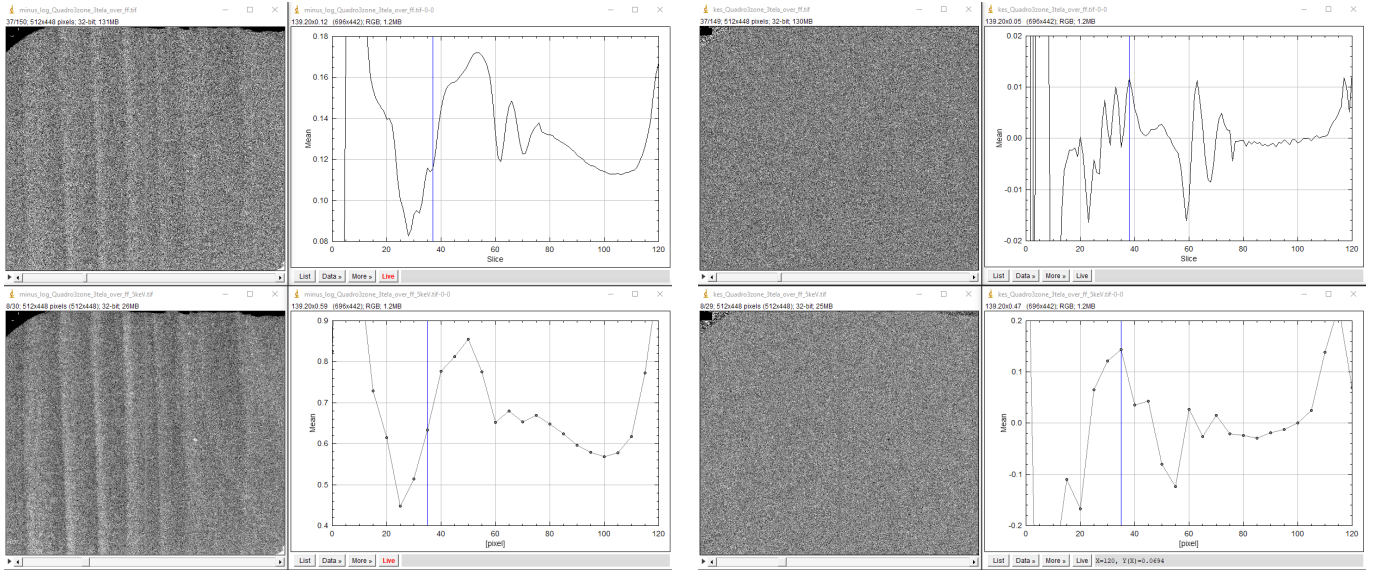
Figure 18: $-\ln(I/I_0)$ and Kes images for ROI 2, 60 kVp

Wood grain and canvas texture can be clearly recognised in total attenuation images, but no evident distinction between the green and the red pigments can be assessed. From Kes analyses a K-edge around 37 keV emerges, and a smaller peak around 26 keV , as seen in Figure 18 (b), could be representative of the presence of cadmium. Pigments are probably too thin to allow a verification of such hypothesis without further analyses. Moreover, no expected background or hidden drawing is identifiable.

Zone 3, 120 kVp



Figure 19: Zone “3” is analysed at 120 kVp and also contains unpainted canvas



(a) Total attenuation, $kVp = 120$ keV , with binwidth 1 keV (upper picture) and 5 keV (lower picture)

(b) K-edge subtraction, $kVp = 120$ keV , with binwidth 1 keV (upper picture) and 5 keV (lower picture)

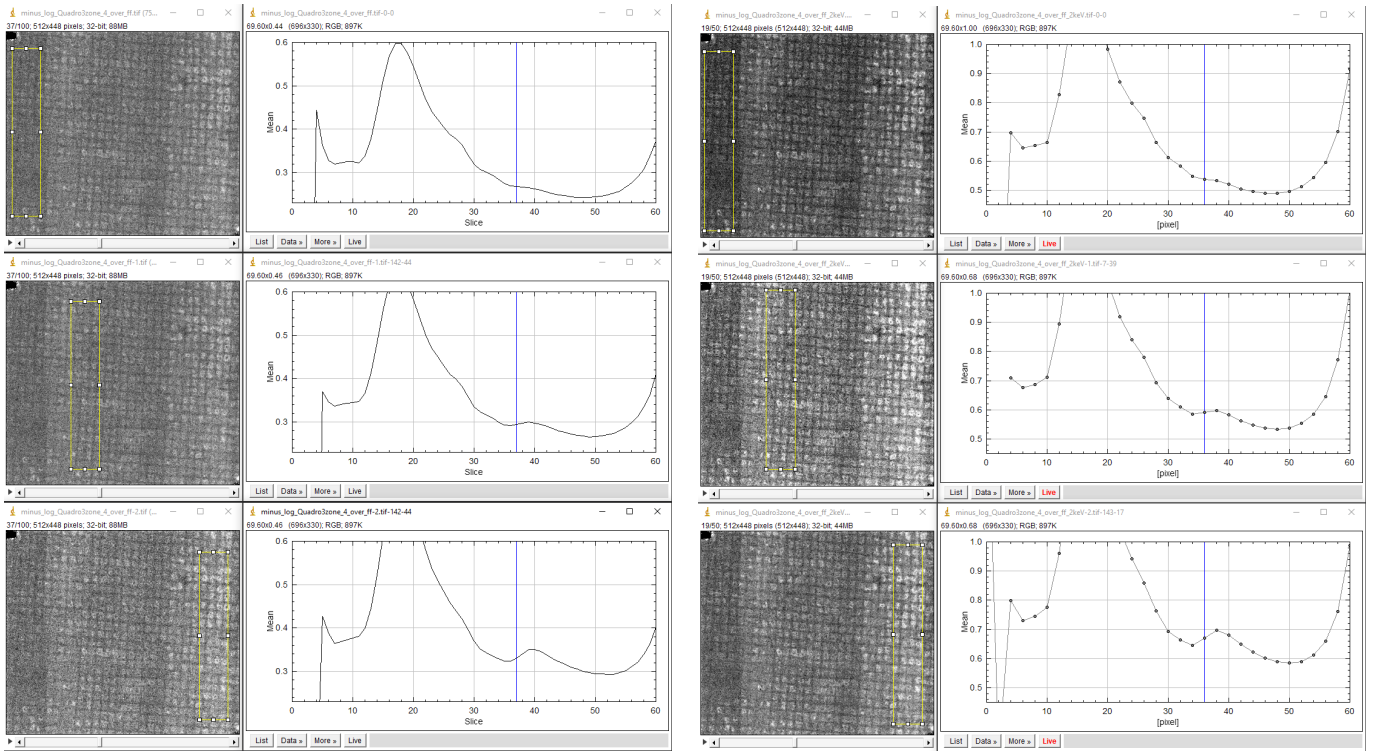
Figure 20: $-\ln(I/I_0)$ and Kes images for ROI 3, 120 kVp

Wood grain and barium K-edge can be recognised, but pigments are apparently too thin to even be distinguished from unpainted canvas (there is no evident distinction between signals taken from a ROI that is expected to contain only the assumed barium sulphate and a ROI that is expected to also contain golden and green pigments). As already stated, gold’s K-edge, expected around 80 keV , does not appear at all.

Zone 4, 60 kVp



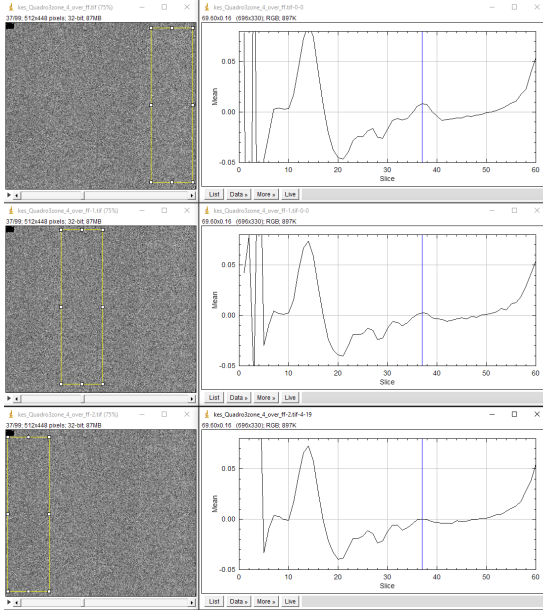
Figure 21: Zone “4” is analysed at 60 kVp



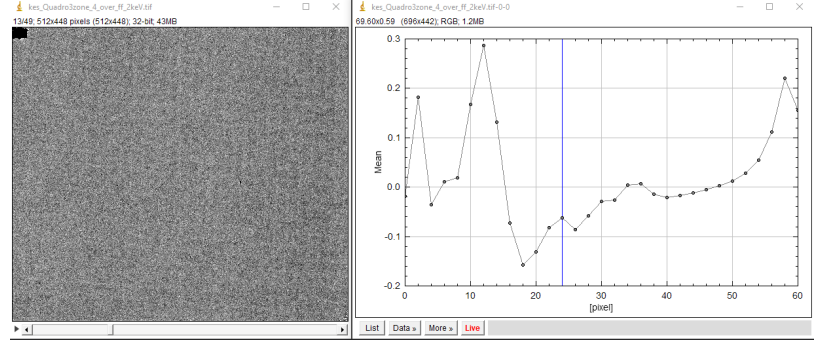
(a) Total attenuation, $kVp = 60$ keV , with binwidth 1 keV

(b) Total attenuation, $kVp = 60$ keV , with binwidth 2 keV

Figure 22: $-\ln(I/I_0)$ images for ROI 4, 60 kVp



(a) Kes image, $kVp = 60 \text{ keV}$, binwidth 1 keV



(b) Kes image, $kVp = 60 \text{ keV}$, with binwidth 2 keV

Figure 23: K-edge subtraction images for ROI 4, 60 kVp

Analyses on this zone confirm what expected: the yellow pigment is the most attenuating one, the blue pigment is the least attenuating one. As a matter of fact, barium's K-edge is clearly visible when focusing on a blue ROI as seen in Figure 22 (a) and Figure 22 (b), bottom pictures, whereas it is almost completely unidentifiable when focusing on a yellow ROI, as it can be seen in the top pictures of Figure 22. Consistently, Kes images display a gradually smaller peak around $E = 37 \text{ keV}$ when moving from the blue pigment (Figure 23 (a), top picture) to the red pigment and then the yellow pigment (Figure 23 (a), bottom picture).

As shown in Figure 23 (b), a less intense peak appears around $E = 26 \text{ keV}$, which could be attributable to cadmium presence. Nevertheless, as it can be noticed in Figure 23 (a), top picture, such peak also appears on the blue pigment ROI: since cadmium is not typically associated to blue colours, the possibility of the peak being caused by noise or artifacts should not be excluded. On the other hand, the blue pigment seems to be superimposed to a substrate of red pigment, which is expected to be thicker than the first one: hence, the 26 keV peak might actually be the proof of cadmium presence in the red pigment.

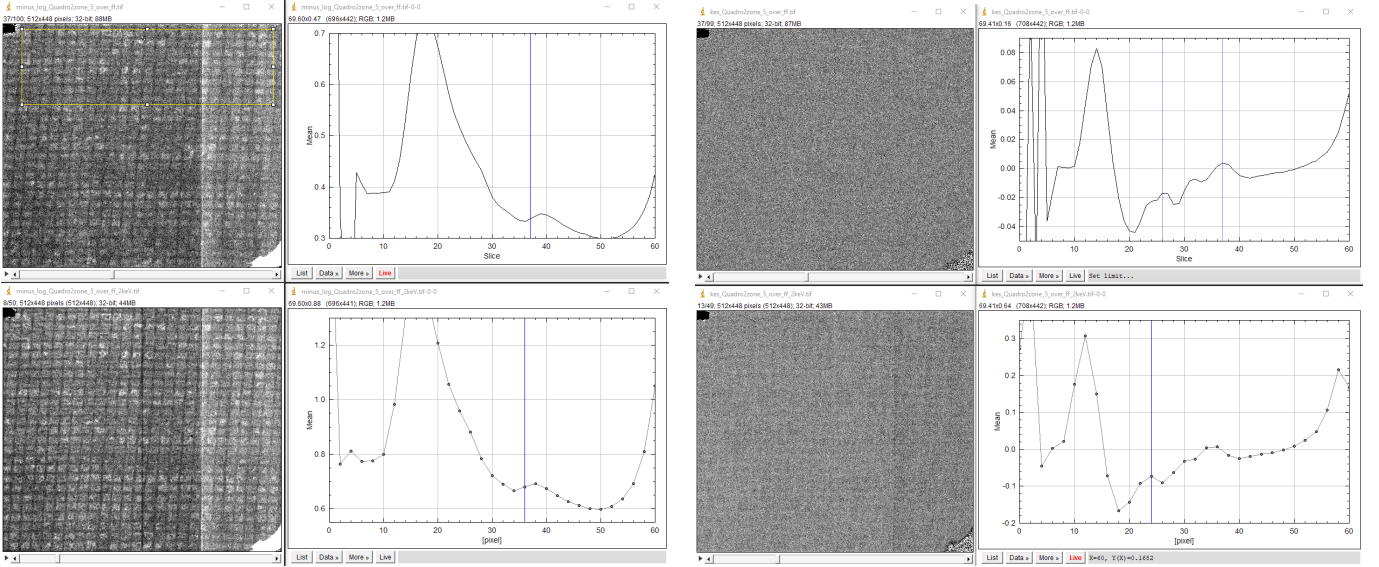
It is not possible to conclude precisely without further analyses on the panel.

Zone 5, 60 kVp

Results on this zone also seem to indicate the presence of barium, as its K-edge appears both globally (as seen in Figure 25 (a), bottom picture) and on a single ROI (as seen in Figure 25 (a), top picture). The two different pigments cannot be distinguished with such imaging analyses, whereas the canvas texture is once again clearly visible. As it can be noticed from K-edge subtraction analyses, a peak around 26 keV arises in addition to the one around 37 keV , which could be characteristic of the presence of cadmium (especially concerning the red pigment). Nevertheless, results are insufficient to univocally identify pigments.



Figure 24: Zone “5” is analysed at 60 kVp



(a) Total attenuation, $kVp = 60 \text{ keV}$, with binwidth 1 keV (upper picture) and 2 keV (lower picture)

(b) K-edge subtraction, $kVp = 60 \text{ keV}$, with binwidth 1 keV (upper picture) and 2 keV (lower picture)

Figure 25: $-\ln(I/I_0)$ and Kes images for ROI 5, 60 kVp

Conclusions

Amptek and Timepix4 analyses on the considered painted wood panel highlight a limitation in K-edge analysis and X-ray imaging techniques: if wood width is much larger than pigments thickness, no precise information can be extracted from the measured signals. In this case, wood (with canvas) could be up to 10000 times thicker than pure pigments (as the wood panel is roughly $\sim 2 \text{ cm}$ wide and pigments are expected to be thinner than $10 \mu\text{m}$). Consistently, imaging slices (from Timepix4 measurements) always display either wood grain or canvas texture (or both), whereas they rarely allow to visually distinguish between different pigments (among six analysed zones, only ROI 4 and ROI 0 show actually different sub-zones).

It is not to be excluded that some of the analysed pigments might contain cadmium (namely, yellow, red and green colours), although signals that could be related to cadmium K-edge only appear in Timepix4 measurements. All results lead to the conclusion that the white pigment spread all over the panel as a substrate contains barium and possibly zinc. A hypothesis could be formulated concerning this matter: lithopone is a white powder made of co-precipitated barium

sulphate and zinc sulphide, which is known to be used for preliminary coverage in painting techniques such as oil, watercolor and tempera. Nevertheless, more measurements with different analytical techniques should be performed to verify such hypothesis and gain more information on the present pigments.

Links

- [1] Amptek detectors documentation and specifications [here](#)
- [2] Timepix4 specifications [here](#)
- [3] Details about previous analyses [here](#)
- [4] Tables of linear attenuation coefficients and material densities from NIST [here](#)
- [5] Material K-edges can be found in the X-ray data booklet [here](#)